Framework of this course

❖ Introduction (L1)

❖ Section I Chemical Measurements

1.1 Basic knowledge (L2)

1.2 Error and estimation (L3&L4)

1.3 Linear regression (L5)

❖ Section II Chemical Equilibrium

2.1 pH and buffer solutions (L6)

2.2 Titrations (L7&L8)

2.3 Potentiometry in titrations (L9)

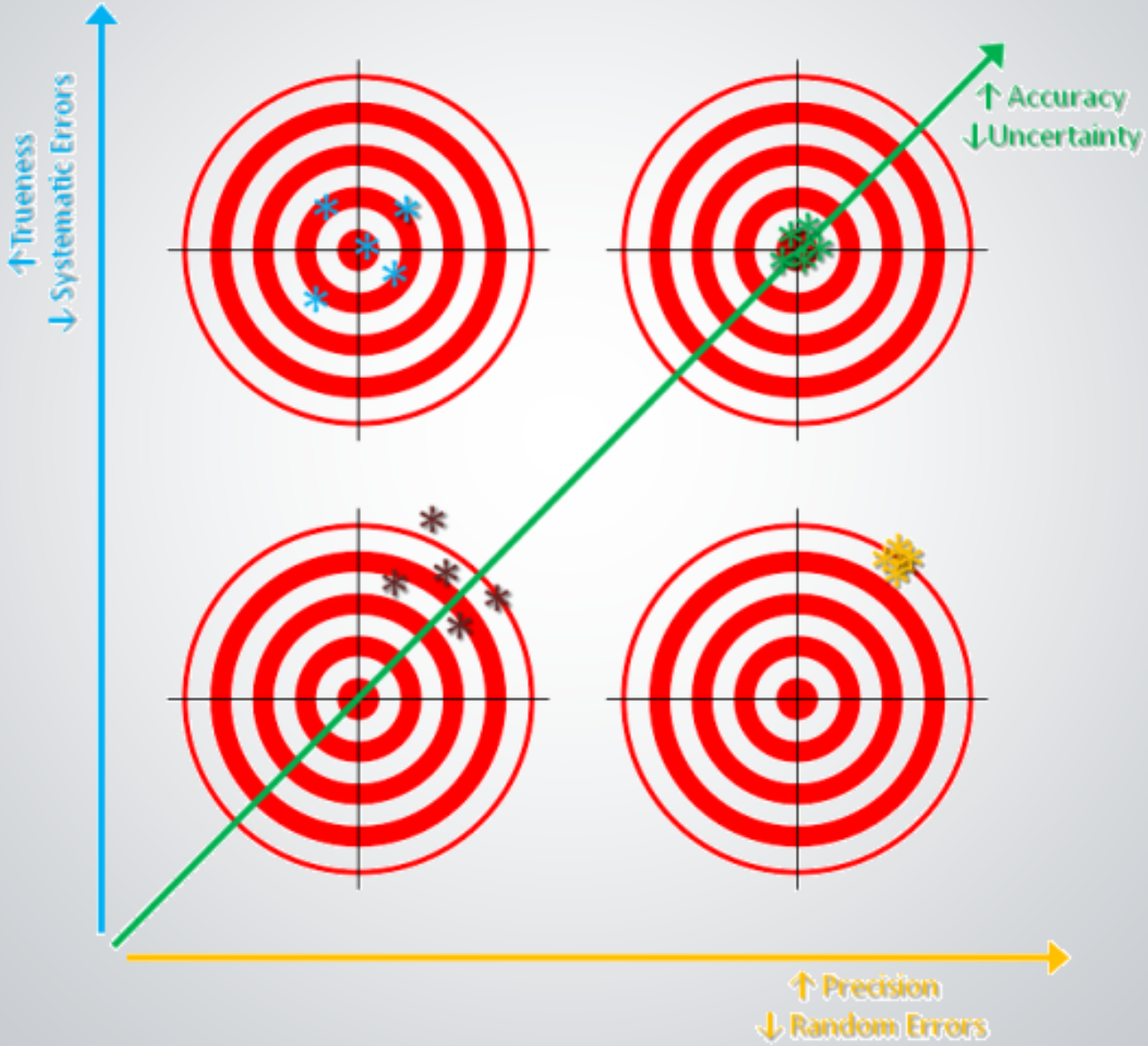
2.4 Gran plot in titrations (L10)

2.5 Electroanalysis (L11)

2.6 gravimetric analysis (L12)

❖ Summary (L13)

Picture showing the most accepted definitions of Accuracy, Trueness and Precision



1.2 Experimental errors & their estimation

Overview

1.2-1 Precision and Accuracy

1.2-2 Significant figure

Keep S.F. in arithmetic

1.2-3 Experimental error

Propagation of errors

1.2-4 Standard deviation

1.2-5 Presentation of results

Targets

1. To know how to keep the significant figures;
2. To distinct between various kinds of error;
3. To calculate “propagation of errors”;
4. To estimate the experimental random error;
5. To know how to report experimental results;

- The characteristics of Random errors

- Also called indeterminate error, arises from uncontrolled variable in the measurement;
- Has an equal chance of being positive and negative, which can not be corrected;
- From the result of statistics.

Deviation in random error

Deviation:

The difference between the value (x_i) in one set and some fixed value, usually the mean ($\bar{x}$)

Individual deviation: $d_i = x_i - \bar{x}$

Average deviation: $\bar{d} = \frac{1}{n}\sum_{i=1}^{n}\left|x_i - \bar{x}\right|$

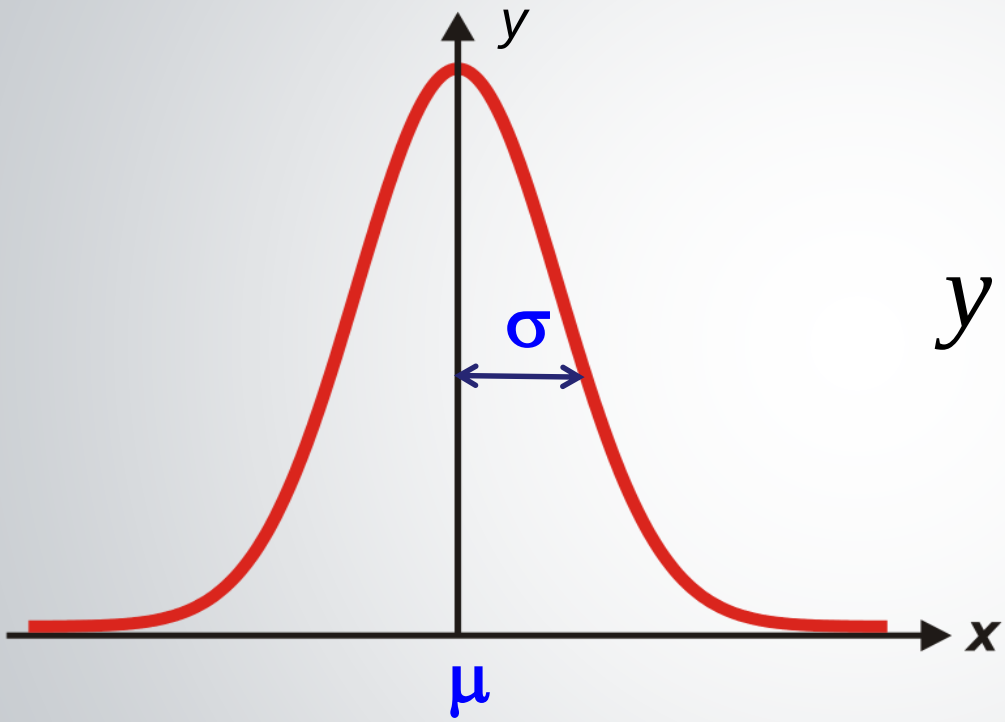
1.2-4 Standard Deviation (SD, *s*)

SD: Measure is used to quantify the amount of variation (dispersion) for a set of data in statistics.

$$s = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{n - 1}}$$

SD is the measure for precision.

Most random error is Normal (Gaussian) distribution



$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\int_{-\infty}^{\infty} y\,dx = 1$$

μ = function center

σ = function width

Normal (Gaussian) distribution

Distribution of

measurement results

Distribution of the

population

Results:

- Average: the cow's weight

- Distribution width: instrument quality

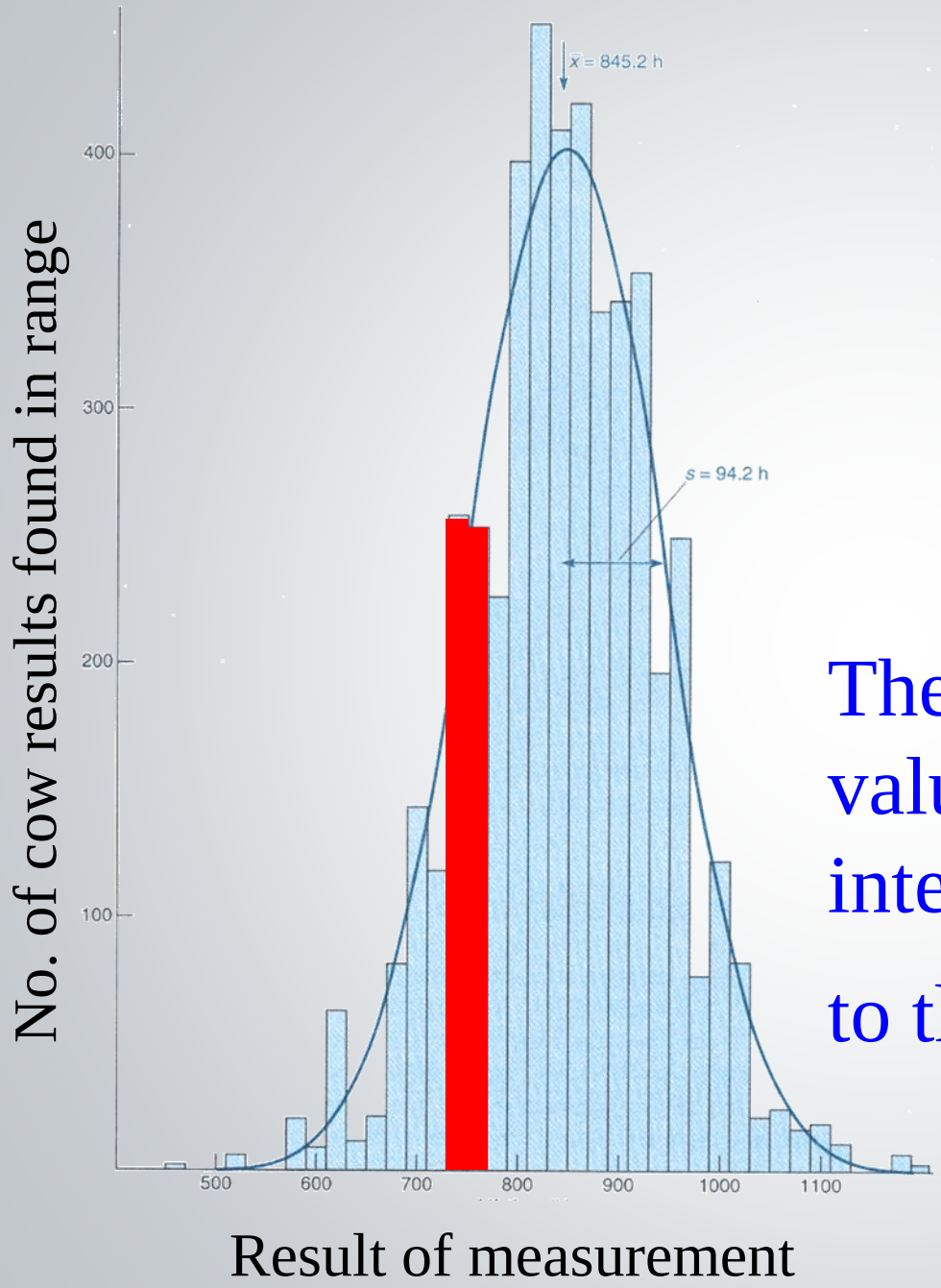
Results:

- The average cows weight μ

- Distribution of weight of cows

Normal distribution of experimental results

No. of cows results found in range	Range
3	690 – 710
6	710 – 730
5	730 – 750
9	750 – 770
16	770 – 790
18	790 – 810
25	810 – 830
16	830 – 850
14	850 – 870
7	870 - 890



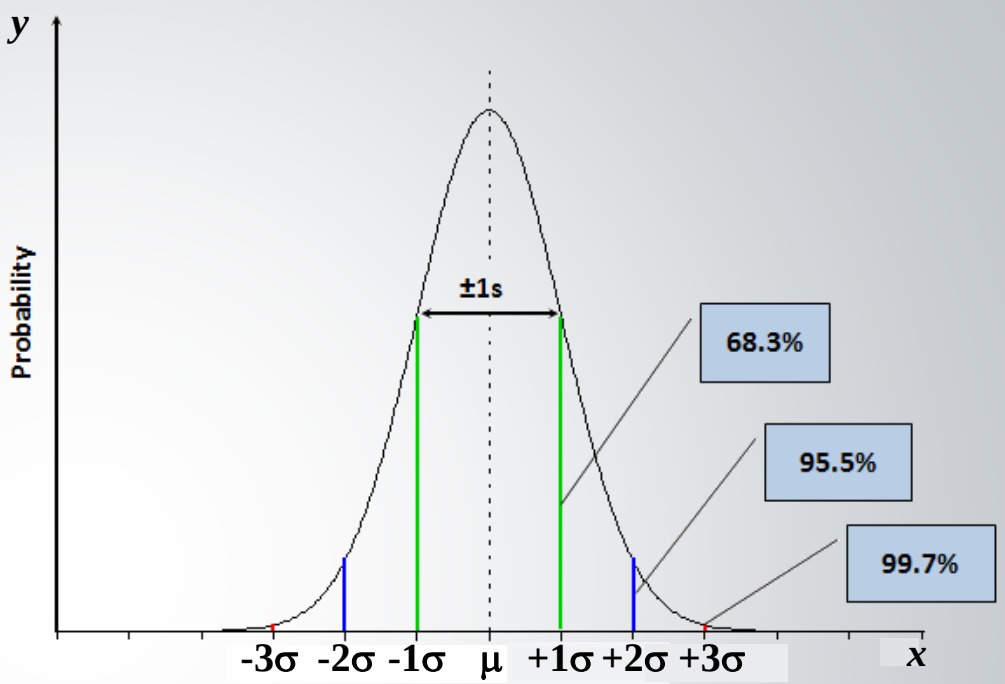
Q: What is the probability to obtain a result in the range 750 to 770?
A: Proportional to the column area.

Q: What is the probability to obtain a result in the range 730 to 770?
A: Proportional to the area of the two columns.

The probability of observed value within a certain intervals is proportional to the area of that interval.

Gaussian curve

$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



The area under the bell-shaped curve

$\mu \pm 1\sigma$	**68.3%(~2/3)**
$\mu \pm 2\sigma$	**95.5%**
$\mu \pm 3\sigma$	**99.7%**

What is the probability that a measurement result is exactly μ?

Estimation of μ and σ from a finite no. of measurements

Estimation of μ:

$$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\bar{x} = \frac{\sum_i x_i}{N}$$

$$\lim_{N \to \infty} \bar{x} = \mu$$

The no. for measurement is N

The population mean is μ

Estimation of σ:

$$\sigma = \sqrt{\frac{\sum_i (x_i - \mu)^2}{N}}$$

If μ is known and only the distribution-width is of interest

If average is calculated from data in finite measurements , only N-1 independent deviations

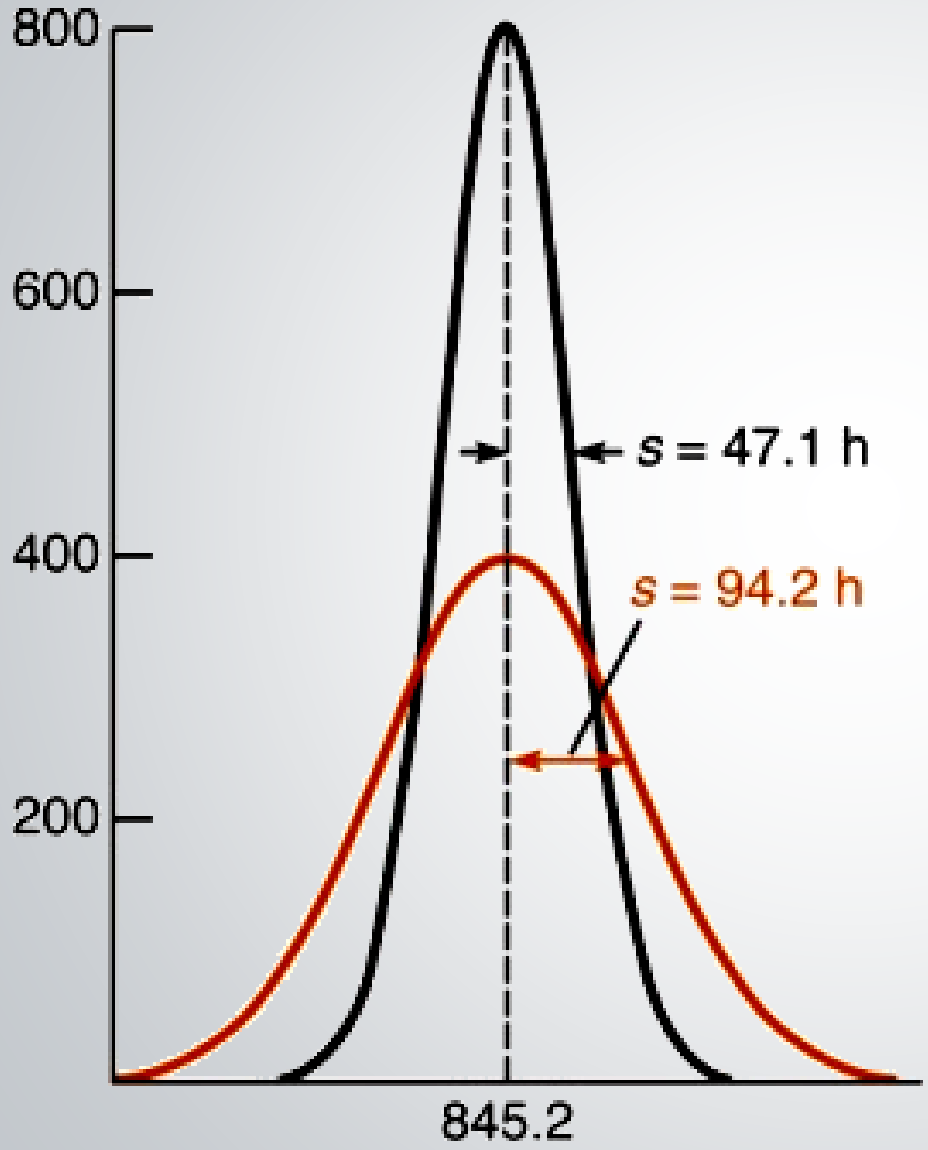
$$s = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{N-1}}$$

N-1 is degree of freedom

$$\lim_{N \to \infty} s = \sigma$$

The population SD is σ

The difference in *s* on the curve



Same average with small *s*:

- **More precise for the results**
- **The probability of observed value is closer to the mean**
- **The data are more clustered**

Note:

- Measurement results would not affected by the measurement itself should be assumed;

- For a finite set of data, we approximate μ by $\bar{x}$; σ by s (which is the measure of uncertainty)

Relative Standard Deviation (RSD)

$$RSD = \frac{s}{\bar{x}} \times 100\%$$

RSD also called coefficient of variation

Find the average $\bar{x}$, standard deviation (SD), and coefficient of variation RSD for a set of number: 821, 783, 834, 855

The average is

$$\bar{x} = \frac{821+783+834+855}{4} = 823._2$$

The standard deviation/SD is

$$s = \sqrt{\frac{(821-823.2)^2+(783-823.2)^2+(834-823.2)^2+(855-823.2)^2}{4-1}}$$

$$= 30._3$$

The coefficient of variation/RSD is

$$\text{RSD} = \frac{s}{\bar{x}} \times 100\% = \frac{30._3}{823._2} 100\% = 3.7\% \approx 4\%$$

Standard deviation of the mean (S_m)

N measurements were performed.

The <u>mean</u> and the SD of the measurements are:

$$\bar{x} = \frac{\sum_i x_i}{N} \qquad s = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{N-1}}$$

Question:

- What is the standard deviation of the means?

Meaning:

- Many sets of measurements are performed and the mean is calculated for each set.
- The SD of these means is calculated for estimating the width of the distribution of the means.

Standard deviation of the mean (S_m)

set-1	$97, 99, 96, \cdots, 96$
set-2	$99, 98, 97, \cdots, 95$
$\vdots$	
set-m	$95, 96, 97, \cdots, 94$

N measurements in total

$$\left.\begin{aligned} \bar{x}_1 &= 96 \\ \bar{x}_2 &= 97 \\ &\\ \bar{x}_M &= 95 \end{aligned}\right\} S_m = \sqrt{\frac{\sum_i (\bar{x}_i - \langle \bar{x} \rangle)^2}{M-1}}$$

Mean of the means

No. of sets

Calculation of S_m directly from the N (N>1) measurements:

$$S_m = \frac{S}{\sqrt{N}}$$

$$S_m < S$$

How the Standard Deviation (of mean) changes with increasing the no. of measurements?

Standard deviation $$S = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{N-1}}$$

Standard deviation of the mean S_m with N measurements $$S_m = \frac{S}{\sqrt{N}}$$

S_m decreases by a factor of $\frac{1}{\sqrt{N}}$ through making N repeated measurements

Standard deviation of the mean (S_m)

First day	Second day	Third day
335	324	340
337	341	327
330	331	333
337	334	335
320	331	345
331.8	332.2	336

$$S_m'(15) = \frac{s(15)}{\sqrt{15}} = \frac{6.54}{\sqrt{15}} = 1.69$$

$$S_m(3) = 2.32$$

First day	Second day	Third day	Fourth day	Fifth day
335	337	330	337	320
324	341	331	334	331
340	327	333	335	345
333	335	331.3	335.3	332

$$S_m'(15) = \frac{s(15)}{\sqrt{15}} = 1.69$$

$$S_m(5) = 1.78$$

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<u>Experimental result</u> is of interest:

- Mean
- SD of the mean
- No. of measurements

$\bar{x} = 0.15$ M

$s_m = \frac{0.06}{\sqrt{6}}$ M = 0.02 M

N = 6

$x = (0.15 \pm 0.02)$ M, $N = 6$

<u>Instrument</u> or <u>method</u> is of interest:

- SD of measurements.
- No. of measurements.

s = 0.02 M

N = 6

($\bar{x} = 0.15$ M)

(M is mol/L)